



CENTRE FOR ADVANCED STUDIES

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY

Jankipuram Vistar, Lucknow, Uttar Pradesh 226031



Artificial Intelligence Lab

An Artificial Intelligence (AI) Lab with high-performance Graphics Processing Units (GPU) server is set up at Centre for Advanced Studies, Dr. A. P. J. Abdul Kalam Technical University, Lucknow, for enabling academic & research institutions for high-class research, course/project work, job-oriented hands-on training in the area of artificial intelligence with a focus on deep learning, machine learning, data science and analytics.

Highlights:

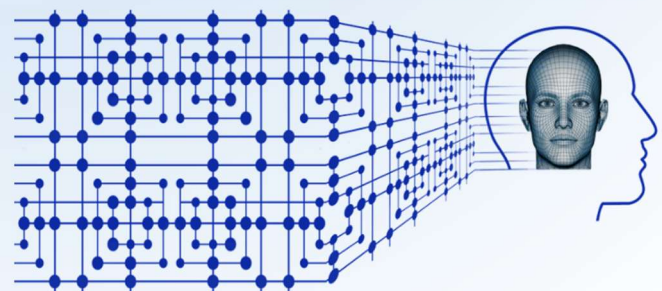
- First Artificial Intelligence Lab in Uttar Pradesh equipped with NVIDIA DGX-2 Servers and Top-5 in country.
- DGX-2 is the world's first 2 petaFLOPS system.
- The System is one of the World's most powerful Deep Learning processing system for the most complex AI Challenges.
- Two DGX-2 GPU enabled Servers are installed which will provide massive parallelization using CUDA technology.
- Capability to train complex models using Multi GPU and provides high levels of inter-GPU bandwidth.
- Will help to train developers, data scientists, and researchers to use deep learning and accelerated computing to solve real-world problems across a wide range of domains in fast time.
- About 2 Lakh computing cores working in parallel to do AI Jobs
- 2 GPU servers performs computations faster than 1000 CPU only servers.
- Consumes only 1/100th of power to deliver 5PF performance compared to traditional computing environment
- DGX-2 has the ability to train ResNet-50 would require the equivalent of 300 servers with dual Intel Xeon Gold CPUs.
- Will reduce carbon footprint by 1/50th compare to traditional computing environment



Computing Power
5 PetaFLOPS

(5 X 10¹⁵ operations per Second)

Built on NVIDIA DGX2 Systems, the new lab Amongst the Top 5 AI Labs in education domain in India and the **first one in Uttar Pradesh**





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Fig 1. DGX-2 GPU Server installed in the institute

Today's college and university students are increasingly tech-savvy, bringing a growing number and variety of personal devices with them to campuses. They also bring expectations of using those devices to support their classwork, with access to the same calibre of tools and graphics-intensive applications as they experience personally, day to day. Educational institutions struggle to keep up with the ever-changing technology needs. They want to offer new ways of learning. Special software and hardware requirements bring students into computer labs to complete their work, adding challenges of scheduling, capacity, and maintenance. These issues, plus the growing interest in remote access options, are causing educational institutions to look for technology solutions that are outside the traditional classroom and computer labs.

The concept of Artificial Intelligence (AI) was first coined in 1955, by Stanford Professor John McCarthy, as the science and engineering of making intelligent machines. In other words, it is a branch of Computer Science, in which a machine mimics the cognitive functions that are associated with the human mind – such as learning and problem solving. Some famous examples include high-level strategic gameplay (e.g. when Google's DeepMind beat the world's best Go player) and autonomous cars. For machines to act and react like humans, they need access to sufficient information from the real world. To be able to initiate common sense, reasoning and problem-solving power, AI needs knowledge engineering – which provides data about objects, categories, properties etc.

Artificial intelligence (AI) is enabling machines and systems to simulate human cognitive functions based on computer science, mathematics, psychology, and neuroscience. Every industry, ranging from health to fitness to media to dating apps to finance, have all adopted AI in some capacity to optimize and automate their processes.

The AI revolution is just getting started and promises to permeate every aspect of human life. But it also poses urgent challenges that cut across firms, developers, government, and workers. Both existing and future workforces will need to be re-skilled to bring out the greatest innovations using AI. This needs to be done quickly to keep up with the AI revolution that's reaching unprecedented levels of adoption.

Application of Artificial Intelligence and Data Science AI solutions in many application areas have now increased the acceptance and very high expectations associated with it. AI can be applied in any domain where it can be trained through the vast amounts of content, able to automatically adapt and learn from its mistakes and failures. Repetitive routine actions that requires large Amount of data or knowledge have the highest potential to benefit from AI system. High adoption segments that are currently are getting exponentially accelerated include:

- a) **Medicine & Healthcare:** Classifications for radiology and pathology, cancer cell detection, diabetic grading, drug Discovery.
- b) **Research in all areas of Science and Engineering :** Scientific Data Analysis and Expert Systems
- c) **Customer Service:** Robotic Concierge, Virtual Assistant, Product Recommendation.
- d) **Maintenance:** Failure Prediction, Maintenance Planning, Anomaly Detection.
- e) **Operations:** Resource Planning, Supply Chain & Logistics, Security & Fraud Detection.
- f) **Automated Devices:** Connected Cars, Robotics, and Intelligent Video Analytics.
- g) **Banking, Financial Services, Insurance:** Wealth Management, Portfolio Planning, Risk Assessment, Underwriting.
- h) **Other Areas:** Public Cloud, Internet, Image Classification, Speech Recognition, Language Translation, Sentiment Analysis, Question Answering, Recommendations.

AI Deep Learning is paradigm shifting Technology driving the fourth industrial revolution. It will transform everything. India has a strategic opportunity to be Global Talent provider to the world for Artificial Intelligence. The major objective of development of this lab is to enable the Institute to offer the very best education, training and research facility in Artificial Intelligence powered by GPUs for AI with focus on deep learning, machine learning, data science and analytics. Technology powered lab will enable Academic Institutions to set up an AI Lab, start electives, short terms courses and take up research efforts in AI; deep learning, machine learning, data science and analytics.

The Artificial Intelligence Lab can also help to train developers, data scientists, and researchers to use deep learning and accelerated computing to solve real-world problems across a wide range of domains. With access to GPU-accelerated workstations in the cloud, researchers and students will learn how to train, optimize, and deploy neural networks using the latest deep learning tools, frameworks, and SDKs. They can also learn how to assess, parallelize, optimize, and deploy GPU-accelerated computing applications. GPU computing leverages the parallel processing capabilities of GPU accelerators and enabling software to deliver dramatic increases in performance for scientific, data analytics, engineering, consumer, and enterprise applications.

Many sound research facilities can be developed for bridging domain experts in various disciplines and their computing needs. There is an increasing trend of using graphics processing units (GPUs) for compute intensive tasks, as these processors have large number of independent cores in a single chip. This has led to the development of supercomputers built on GPUs. The programming of such multicore processors has importance in the current high performance computing domain.

Objective of the AI Lab:

- Identify innovative research directions in Artificial Intelligence, Deep Learning, Machine Learning, and Big Data analytics.
- Encouraging students to publish research articles, patents and starting their start-ups in the campus/university.
- Integrating and reasoning with information from disparate data sources by training students, developers, data scientists, and researchers.
- Designing and implementing distributed systems for information exploitation, collaboration and decision making.
- Data-intensive agent-based tools providing quality education and practical skills to the students and faculty.
- Assist in the development of partnerships with Industry regarding Internships, Summer Jobs, Publications and students' placements.
- Establish, refine and implement strategies to take the idea in to students and faculty fraternity.

For AI system to be intelligent and provide meaningful responses it needs to be trained on the appropriate data. Unlike most conventional learning methods, which are considered using shallow structured learning architectures, deep learning refers to machine learning techniques that use supervised and/or unsupervised strategies to automatically learn hierarchical representations in deep architectures for classification. Deep Learning was constrained with two key factors for practical applicability. One was the availability of Big Data. With the explosion of Big Data with Internet growth solving the Data problem, the second issue was that even with Big Data availability, to get the compute power required to harvest valuable knowledge from Big Data. The large and rapidly growing body of information hidden in the unprecedented volumes of non-traditional data requires both the development of advanced technologies and interdisciplinary teams working in close collaboration. The success of an AI solution is not only dependent on the accuracy of its response. But also on the time taken to (re)train, develop, and scale accordingly. A reduction in time required to train a model will fuel the growth of AI solutions that are released into the market. Several options are available to help reduce the training time of a model. Of all the techniques to solve this problem practically, using GPU based Deep Learning proved to be the only viable option for Deep Learning. The reason was the wide availability of GPUs that make parallel processing ever faster, cheaper, and more powerful. Compute parallelization is a vital ingredient in Deep Learning. Computing the weights and biases of network layers, iterating through network hyper-parameters, forward propagation, error estimations through gradient descent and other calculations are all essentially parallelizable operations. The recognition of this fact by early researchers and NVIDIA is perhaps the single most impactful development in the advance of DL over the past half-decade.

Applications of Artificial Intelligence in Science and Engineering

1. AI in Healthcare:

- The market value of AI in the health care industry is predicted to reach \$6.6 billion by 2021. The impact of AI in the health care sector is genuinely life-changing. It is driving innovations in clinical operations, drug development, and surgery and data management.
- Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization. Possibility of AI in healthcare is to develop a clinical decision support system for disease prevention that can give the physician a warning when there's a patient at risk of having some disease.
- Another possibility is designing a digitalized device that can find disease and recommend medicines for common diseases. Some of the revolutionizing applications of AI in healthcare are explained as follows:
- **Clinical Diagnosis:** AI algorithms diagnose diseases faster and more accurately than doctors. They are particularly successful in detecting diseases from image-based test results. Effective treatment of cancer heavily depends on early detection and pre-emptive measures. AI algorithms can scan and analyse biopsy images and MRI scans 1,000 times faster than doctors. The algorithms can diagnose with more than 87% accuracy rate.
- **Precision Medication:** Precision medication refers to dispensing the correct treatment depending on the patient's characteristics and behaviour. The patient's data is used in determining the most effective medication. The data includes treatment history, restrictions, hereditary traits and lifestyle. AI-powered data management systems seamlessly store and organize large amounts of data to draw meaningful conclusions and predictions.
- **Drug Discovery:** Drug development is a tedious venture that may take years and thousands of failed attempts. It can cost medical researchers billions of dollars in the process. The idea is to build a drug discovery program using deep learning and AI. The results are already paying off. Intelligent simulations of better cures are possible through analysis of the existing medicine, patients and pathogens.
- **Personal Health Assistants:** Over the last few years, we have seen mobile apps, wearables, and discrete monitors that continually collect data and check the vitals. These gadgets use the data to make recommendations. These assistants can inform about any irregularities in health. Most of these devices store data locally or online. The data can be retrieved and used by medical practitioners as a medical report.

2. AI in Data Security

- The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure.
- Some examples such as AEG bot, AI2 Platform, are used to determine software bug and cyber-attacks in a better way.

3. AI in Robotics

- Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task, but AI can be used to create intelligent robots which can perform tasks with their own experiences without pre-programmed.

- Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as Erica and Sophia has been developed which can talk and behave like humans.

4. AI in Agriculture

- Agriculture is applying AI as agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be very helpful for farmers.
- Possibility is to develop a robot which uses computer vision technologies like object detection to monitor and precisely spray weedicide on plants to avoid damaging of plants.
- Another possibility of AI is to identify potential defects and nutrient deficiencies in the soil through images.

5. AI in Drug Development

- AI is widely used in pharmaceutical company for developing drugs.
- Possibility of AI in drug discovery is to find relationship in between disease, target and drug, to find the hidden connections and new insights in diverse, linked datasets, retrieve and visualize relevant results from multiple biomedical data sources, Design protein drugs through machine learning, Automate evaluating the impact of drug combinations on cancer cells.

6. AI in education:

- AI can automate grading so that the tutor can have more time to teach. AI chatbot can communicate with students as a teaching assistant.
- Possibility of AI in the future can be work as a personal virtual tutor for students, which will be accessible easily at any time and any place.
- Potential use cases include augmenting and enhancing the learning experience through personalised learning, automating and expediting administrative tasks, and predicting the need for student intervention to reduce dropouts or recommend vocational training.
- **Differentiated and individualized learning:** As AI gets more sophisticated, it might be possible for a machine to read the expression that passes on a student's face that indicates they are struggling to grasp a subject and will modify a lesson to respond to that. The idea of customizing curriculum for every student's needs is a possible future application of AI in education.
- **Automation of admin tasks:** An educator spends a tremendous amount of time grading homework and tests. AI can step in and make quick work out of these tasks. There is much potential for AI to create more efficient enrolment and admissions processes.
- **Universal access for all students:** Artificial intelligence tools can help make global classrooms available to all including those who speak different languages or who might have visual or hearing impairments. So that the education can reach to every child in the country.

7. AI in Social Media

- Social Media sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyse lots of data to identify the latest trends, hashtag, and requirement of different users.
- In social media platforms like Facebook, AI is used for face verification wherein machine learning and deep learning concepts are used to detect facial features and tag your friends. Deep Learning is used to extract every minute detail from an image by using a bunch of deep neural networks. On the other hand, Machine learning algorithms are used to design your feed based on your interests.
- AI can be utilized to make use of Machine Learning, Deep Learning, and Natural language processing to filter out violent and unwanted content. AI can also be used to make data shared on social media safer. Analyse the account on the basis of content shared by the user.

8. AI in Space Exploration

- Artificial Intelligence is also being used for NASA's next rover mission to Mars, the Mars 2020 Rover. The AEGIS, which is an AI-based Mars rover is already on the red planet. The rover is responsible for autonomous targeting of cameras in order to perform investigations on Mars. After rigorous research, astronomers used Artificial Intelligence to sift through years of data obtained by the Kepler telescope in order to identify a distant eight-planet solar system.
- Space expeditions and discoveries always require analysing vast amounts of data. Artificial Intelligence and Machine learning is the best way to handle and process data on this scale.

9. AI in Autonomous Vehicles

- For the longest time, self-driving cars have been a buzzword in the AI industry. The development of autonomous vehicles will definitely revolutionaries the transport system.
- The AI system can collect data from the vehicle's radar, cameras, GPS, and cloud services to produce control signals that operate the vehicle. Advanced Deep Learning algorithms can accurately predict what objects in the vehicle's vicinity are likely to do.
- Artificial Intelligence implements computer vision, image detection and deep learning to build cars that can automatically detect objects and drive around without human intervention.

10. AI in Banking

- AI in banking is growing faster than you thought! A lot of banks have already adopted AI-based systems to provide customer support, detect anomalies and credit card frauds.
- For example HDFC Bank has developed an AI-based chatbot called **EVA** (Electronic Virtual Assistant), built by Bengaluru-based Senseforth AI Research. It addressed over 3 million customer queries, interacted with over half a million unique users, and held over a million conversations. Eva can collect knowledge from thousands of sources and provide simple answers in less than 0.4 seconds.
- AI and Deep Learning can be used to detect fraudulent transaction patterns and prevent card fraud.

11. AI in Chatbots

- Virtual assistants have become a very common technology. A few examples include Siri, Cortana, which are gaining popularity because of the user experience they provide.
- Amazon's Echo is an example of how Artificial Intelligence can be used to translate human language into desirable actions. Another example is the newly released Google's virtual assistant called Google Duplex that has astonished millions of people.
- Natural language processing and machine learning algorithms can be used to process human language and perform tasks on the basis of its user's mood, after reading human face emotions.

12. AI in Artificial Creativity

- An AI-based system called MuseNet can now compose classical music that echoes the classical legends, Bach and Mozart. MuseNet is a deep neural network that is capable of generating 4-minute musical compositions with 10 different instruments and can combine styles from country to Mozart to the Beatles.
- Artificially Intelligent machine can be used to create music and art using a natural language generation platform that can transform your data into insightful narratives.
- Using machine learning algorithms and feeding those algorithms great quantities of data, IT and security experts can teach the AI solution to monitor behaviour, detect anomalies, adapt and respond to threats and issue alerts. AI has quickly become a key component in a business' cybersecurity infrastructure.

13. AI in Entertainment

- AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show the recommendations for programs or shows.
- AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business.
- Possibility of AI is to help shoppers to discover associated products with recommended size, color, or even brand.

14. AI in Defence

- Artificial intelligence is very helpful in providing predictive results from the analysis of the collected data. That's the main reason behind AI why it is increasingly becoming an integral part of the defence and modern warfare systems. AI is being used to enhance military intelligence by compiling it with the results given by the AI systems. Some of the major applications of AI systems in defence mechanisms are explained below in detail.
- **Recognition of target:** The most vital aspect of any kind of defence activity is the recognition of potential targets. AI systems are widely being used to compile data from previous attacks and activities to identify the hotspots for enemy activities. The information stored in the database can be used effectively to identify the targets and locations which can pose a major threat.

- **Medical service on the battlefield:** Medical backup is very crucial for the nation's defence. It is known fact that it is very difficult to provide medical support on the battlefield. For this reason, there is an opportunity to develop prototypes that are fully AI controlled medical assistants to provide medical support for the soldiers.
- **Cybersecurity:** In today's digital era where almost every data from every transaction is available on digital platforms. This data often includes sensitive and confidential information about the nation's defence systems so the concept of defence data security becomes an important factor. AI systems can learn the patterns of the authorised personnel and differentiate that from the behaviour of any malicious software or any attempt to access confidential information from an unreliable source. AI systems automatically detect threats and prevent any cyber-attack at the primary stage.
- **Surveillance:** It is important for accumulating the necessary information about the enemies and devise the defensive strategy accordingly. However, performing surveillance operations can be very dangerous as often one has to venture into the enemy lines to gather necessary intelligence. AI systems have given us a major benefit in this avenue by allowing us to create un-manned surveillance systems that can enter hostile areas and collect the required data. They can collect and process more data compared to their human counterparts and are also more effective in identifying threats.
- **Transportation and Logistics:** Logistics and transportation services are very crucial for any defence system as the supply of essential equipment for the soldiers should always be kept un-interrupted for the sake of security. AI is being incorporated increasingly in transportation systems to enhance the security of various military transportation and defence equipment. The AI system can be programmed to detect any kind of anomalous behaviour and take the necessary measures to tackle them. AI is very useful for providing logistics as they can give an accurate reading of faults and failures in the various components which require immediate attention. This system can be used to perform predictive maintenance and reduce errors in reasonable time consumption.

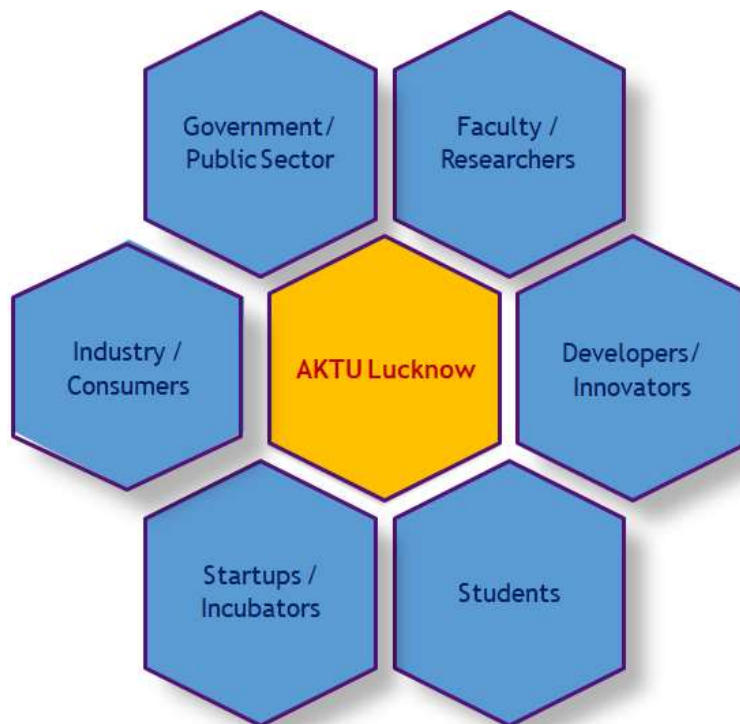
16 AI in Environment

- Leveraging AI and IoT for environmental sustainability can help maximize our current efforts for environmental protection. Some of the potential areas where the AI can revolutionize our environmental protections efforts are as follows:
- **AI for renewable energy technology:** Renewable energy technologies are already useful in cutting down environmentally hazardous activities. Applying AI based IoT machinery to existing practices can help maximize the benefits of renewable energy methods. It will help make renewable energy solutions like solar panels and wind turbines more efficient and cost-effective. These grids give energy providers a better understanding of power usage and provide the ability to make real-time adjustments for efficiency. This will lower our dependence on fossil fuels, hopefully eliminating their need completely.
- **AI for agricultural sustainability:** Soil pollution is a major issue faced today due to population growth, intensive farming, and other activities. Application of AI and IoT for environmental sustainability includes methods to monitor crops and soil and maximize crop production with a low impact on the environment. Smart monitoring devices and sensors can be attached to crops to monitor their growth constantly. Anomalies can be detected and resolved immediately. Parameters such as hydration, plant nutrition, and diseases can be monitored in real-time.
- **AI for cleaner air:** Air pollution can cause chronic and respiratory problems like asthma and such fatal diseases as lung cancer. IoT and AI can provide tools to monitor air pollution in real-time. The tools can identify sources of air pollution quickly and accurately. For example, the source of a gas leak in an industry can be identified using smart sensors and remedial measures can be applied effectively. Government organizations and commercial enterprises can benefit hugely by using IoT and AI technologies to monitor air purity levels and alert employees if air quality falls below a certain level.
- **AI for water preservation:** Human exploitation of natural resources has had a harmful impact on major water bodies. Machine learning can also help assess the changing conditions that harm the ocean, such as illegal fishing, discharging industrial effluents into water bodies, and illegal solid waste disposal by the local people. IoT and AI can also help improve the water quality by monitoring pollution levels in real-time to determine if any illegal activities are hampering the water quality.

Technical Advantages of the AI Lab

- Students/Research scholars should be able to run their Deep Learning/Machine Learning training workloads on the DGX Servers /Cluster.
- Authentication and Authorization based mechanism for each user group for security, confidentiality and accessibility.
- AI workload scheduling with orchestration tools to handle large no. of jobs.
- High speed interconnect between DGX servers and high speed storage system to manage the requirements of AI workloads in terms of storage performance and capacity.
- DGX servers with 16 GPU & 4 GUP are provisioned for DL models training so that it can span multi GPU as well as multi servers to provide seamless compute resources to feed the model compute requirement..
- Storage with high throughput is provisioned to provide the storage space for training data. Storage is provisioned with a mix of enterprise class high speed SSDs and SAS drives to keep a balance in cost and performance.
- Mellanox Switches are being provided to support the high-speed connectivity (100 Gbps) between storage and DGX.
- NFS shares on high throughput storage is presented to DGX servers. Connectivity between DGX and Storage is provided through Mellanox switches to provide low latency data transfer in order to obtain the expected performance benefits. All DGX connects together on 100gbps speed.

ECOSYSTEM

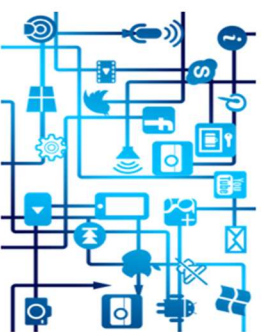




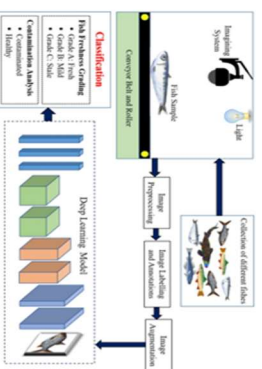
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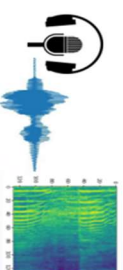
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Networking and Social Media



Fish Freshness and Quality Assessment



Time Series Signals



Face Recognition



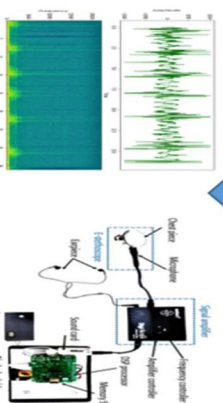
Automatic Navigation



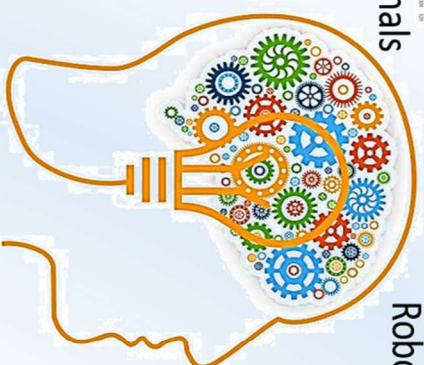
Object Recognition



Document Analysis



Cardiac and Pulmonary Sound Analysis



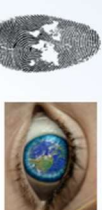
Artificial Intelligence Lab



Robotics



Security



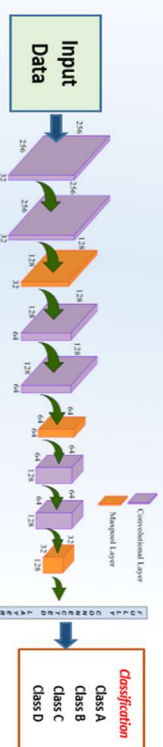
Biometrics



Speech Recognition



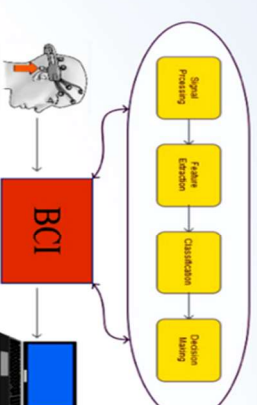
Medical Imaging



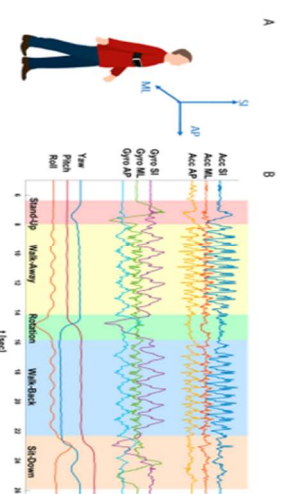
Deep Neural Network



Assistive Device for Visually Impaired People



Brain Computer Interface using EEG Headset



Gait Pattern Analysis

